

Model description: Spiking Temporal Memory (STM) model

1 Network model

Summary	
Populations	excitatory ($\mathcal{E}$), inhibitory ($\mathcal{I}$) and external ($\mathcal{X}$)
Connectivity	• sparse random connectivity between excitatory neurons (plastic) • local recurrent connectivity between excitatory and inhibitory neurons (static)
Neuron model	• excitatory neurons: leaky integrate-and-fire (LIF) with nonlinear input integration (dendritic action potentials) • inhibitory neurons: leaky integrate-and-fire (LIF)
Synapse model	exponential or alpha-shaped postsynaptic currents (PSCs)
Plasticity	spike-timing dependent structural plasticity and weight decay in excitatory to excitatory connections
Input	external spike sources, connected to excitatory neurons

(see legend)

Table 1: Summary of the network model. Parameter values are given in Table Tab. 9.

Populations		
$\mathcal{E} = \cup_{i=k}^M \mathcal{M}_k$	excitatory (E) neurons	N_E
$\mathcal{I}$	inhibitory (I) neurons	N_I
$\mathcal{M}_k$	excitatory neurons in subpopulation k , $\mathcal{M}_k \cap \mathcal{M}_l = \emptyset$ ($\forall k \neq l \in [1, M]$)	n_E

Table 2: Description of the populations. Parameter values are given in Tab. 9

Connectivity		
Source pop-ulation	Target pop-ulation	Pattern
$\mathcal{E}$	$\mathcal{E}$	random; fixed in-degrees $K_i = K_{EE}$, delays $d_{ij} = d_{EE}$, and synaptic time constants $\tau_{ij} = \tau_{EE}$, plastic synaptic weights J_{ij} ($\forall i \in \mathcal{E}, \forall j \in \mathcal{E}$; "EE connections")
$\mathcal{E}$	$\mathcal{I}$	all-to-all; fixed delays $d_{ij} = d_{IE}$, synaptic time constants $\tau_{ij} = \tau_{IE}$, and weights $J_{ij} = J_{IE}$ ($\forall i \in \mathcal{I}, \forall j \in \mathcal{E}$; "IE connections")
$\mathcal{I}$	$\mathcal{E}$	all-to-all; fixed delays $d_{ij} = d_{EI}$, synaptic time constants $\tau_{ij} = \tau_{EI}$, and weights $J_{ij} = J_{EI}$ ($\forall i \in \mathcal{E}, \forall j \in \mathcal{I}$; "EI connections")
$\mathcal{I}$	$\mathcal{I}$	none ("II connections")
all	all	no self-connections ("autapses"), no multiple connections ("multapses")

Table 3: Description of the connectivity. Parameter values are given in Table Tab. 9.

Neuron	
Type	leaky integrate-and-fire (LIF) dynamics
Description	dynamics of membrane potential $V_i(t)$ and spiking activity $s_i(t)$ of neuron i: - emission of the kth spike of neuron i at time t_i^k if $V_i(t_i^k) \geq \theta_i \quad (1)$ with somatic spike threshold θ_i - spike train: $s_i(t) = \sum_k \delta(t - t_i^k)$ - reset and refractoriness: $V_i(t) = V_r \quad \forall k, \forall t \in (t_i^k, t_i^k + \tau_{\text{ref},i}]$ with refractory time $\tau_{\text{ref},i}$ and reset potential V_r - subthreshold dynamics: $\tau_{m,i} \dot{V}_i(t) = -V_i(t) + R_{m,i} I_i(t) \quad \forall k, \forall t \notin (t_i^k, t_i^k + \tau_{\text{ref},i}] \quad (2)$ with membrane resistance $R_{m,i} = \frac{\tau_{m,i}}{C_{m,i}}$, membrane time constant $\tau_{m,i}$, and total synaptic input current $I_i(t)$ (see Tab. 5) - excitatory neurons: $\tau_{m,i} = \tau_{m,E}$, $C_{m,i} = C_m$, $\theta_i = \theta_E$, $\tau_{\text{ref},i} = \tau_{\text{ref},E}$ ($\forall i \in \mathcal{E}$) - inhibitory neurons: $\tau_{m,i} = \tau_{m,I}$, $C_{m,i} = C_m$, $\theta_i = \theta_I$, $\tau_{\text{ref},i} = \tau_{\text{ref},I}$ ($\forall i \in \mathcal{I}$)

Table 4: Description of the neuron model. Parameter values are given in Tab. 9.

Synapse	
Type	continuous, exponential, or alpha-shaped postsynaptic currents (PSCs)
Description	- total synaptic input current excitatory neurons: $I_i(t) = I_{ED,i}(t) + I_{EX,i}(t) + I_{EI,i}(t), \forall i \in \mathcal{E}$ inhibitory neurons: $I_i(t) = I_{IE,i}(t), \forall i \in \mathcal{I}$ (3) with dendritic, inhibitory, excitatory, and external input currents $I_{ED,i}(t)$, $I_{EI,i}(t)$, $I_{IE,i}(t)$, $I_{EX,i}(t)$ evolving according to $I_{ED,i}(t) = \sum_{j \in \mathcal{E}} (\alpha_{ij} * s_j)(t - d_{ij}) \quad (4)$ with $\alpha_{ij}(t) = J_{ij} \frac{e}{\tau_{ED}} t e^{-t/\tau_{ED}} \Theta(t)$ and $\Theta(t) = \begin{cases} 1 & t \geq 0 \\ 0 & \text{else} \end{cases}$ $\tau_{EI} \dot{I}_{EI,i} = -I_{EI,i}(t) + \sum_{j \in \mathcal{I}} J_{ij} s_j(t - d_{ij}) \quad (5)$ $\tau_{IE} \dot{I}_{IE,i} = -I_{IE,i}(t) + \sum_{j \in \mathcal{E}} J_{ij} s_j(t - d_{ij}) \quad (6)$ $I_{EX,i}(t) = I_{S,i}(t) + I_{B,i}(t) \quad (7)$ where $I_{S,i}(t)$ is the stimulus input (see Tab. 7:Input). - suprathreshold dynamics of dendritic currents (dAP generation): - emission of kth dAP of neuron i at time $t_{dAP,i}^k$ if $I_{ED,i}(t_{dAP,i}^k) \geq \theta_{dAP}$ - dAP current plateau: $I_{ED,i}(t) = I_{dAP} \quad \forall k, \forall t \in (t_{dAP,i}^k, t_{dAP,i}^k + \tau_{dAP}) \quad (8)$ with dAP current plateau amplitude I_{dAP}, dAP current duration τ_{dAP}, and dAP activation threshold θ_{dAP} - reset: $I_{ED,i}(t_{dAP,i}^k + \tau_{dAP}) = 0$ ($\forall k$) - reset and refractoriness in response to emission of lth somatic spike of neuron i at time t_i^l: $I_{ED,i}(t) = 0 \quad \forall l, \forall t \in (t_i^l, t_i^l + \tau_{ref,i}) \quad (9)$

Table 5: Description of the synapse model. Parameter values are given in Tab. 9.

Plasticity	
Type	spike-timing dependent structural plasticity and permanence decay
EE synapses	- dynamics of synaptic permanence $P_{ij}(t)$ (maturity) in EE connections during learning: $\forall \{t P_{\min,ij} < P_{ij}(t) < P_{\max}\} :$ $\begin{aligned} \frac{dP_{ij}}{dt} = & P_{\max} \lambda_+ \sum_{\{t_i^*\}'} \delta(t - [t_i^* + d_{EE}]) I_+(x_j(t), t_i^*, \Delta t_{\min}, \Delta t_{\max}) \\ & - P_{\max} \lambda_- \sum_{\{t_j^*\}'} \delta(t - [t_j^* + d_{EE}]) I_-(x_i(t), t_j^*, \Delta t_{\max}) \\ & + (P_{\min,ij} - P_{ij}) \frac{1}{\tau_P} \end{aligned} \quad (10)$ $\forall \{t P_{ij}(t) < P_{\min,ij}\} : \quad P_{ij}(t) = P_{\min,ij}$ $\forall \{t P_{ij}(t) > P_{\max}\} : \quad P_{ij}(t) = P_{\max}$ - with - list of presynaptic spike times $\{t_j^*\}$, - list of postsynaptic spike times $\{t_i^*\}' = \{t_i^* \forall t_j^* : t_i^* - t_j^* + d_{EE} \geq \Delta t_{\min}\}$ - increment functions $I_+(x_j(t), t_i^*, \Delta t_{\min}, \Delta t_{\max}) = R_+(t_i^* - t_j^+ + d_{EE})$ with $R_+(\tau) = \begin{cases} x_j(t) & \Delta t_{\min} < \tau < \Delta t_{\max} \\ (x_j(t) - 1) & \tau < \Delta t_{\min} \\ 0 & \text{else,} \end{cases} \quad (11)$ $I_-(x_i(t), t_j^*, \Delta t_{\max}) = R_-(t_j^* - t_i^- + d_{EE})$ with $R_-(\tau) = \begin{cases} x_i(t) & \tau < \Delta t_{\max} \\ 0 & \text{else,} \end{cases}$ - maximum permanence $P_{\max}$, minimum permanence $P_{\min,ij}$, potentiation and depression rates λ_+, λ_-, decay time constant τ_P, delay d_{EE}, minimum $\Delta t_{\min}$ and maximum $\Delta t_{\max}$ time lags between pairs of pre- and postsynaptic spikes at which synapses are potentiated or depressed, nearest presynaptic spike time t_j^+ preceding t_i^*, nearest postsynaptic spike time t_i^- preceding t_j^*, - spike trace of presynaptic neuron j, evolving according to $\frac{dx_j}{dt} = -\tau_+^{-1} x_j(t) + \sum_{t_j^*} \delta(t - t_j^*)$ with presynaptic spike times t_j^* and potentiation time constant τ_+, - spike trace of postsynaptic neuron i, evolving according to $\frac{dx_i}{dt} = -\tau_-^{-1} x_i(t) + \sum_{t_i^*} \delta(t - t_i^*)$ with postsynaptic spike times t_i^* and depression time constant τ_-. - dynamics of synaptic weights $J_{EE,ij}$ according to $J_{EE,ij} = \begin{cases} J_{\max} & P_{ij} > P_\theta \\ 0 & \text{else,} \end{cases}$ with maximum synaptic weight $J_{\max}$ and synapse maturity threshold P_θ.
all other synapses	non-plastic

Table 6: Description of the plasticity model. Parameter values are given in Tab.9.

Input
• repetitive stimulation of the network using the same set $\mathcal{S} = \{s_1, \dots, s_S\}$ of sequences $s_i = \{\zeta_{i,1}, \zeta_{i,2}, \dots, \zeta_{i,C_i}\}$ of ordered discrete items $\zeta_{i,j}$ with number of sequences S and length C_i of ith sequence • presentation of sequence element $\zeta_{i,j}$ at time $t_{i,j}$ modeled by a single spike $x_k(t) = \delta(t - t_{i,j})$ generated by the corresponding external source x_k • generated current as a response to the presentation of the sequence elements: $\tau_S \dot{I}_{S,i} = -I_{S,i}(t) + \sum_{j \in \mathcal{X}} J_{i,j} x_j(t - d_{ij}) \quad (12)$ • inter-stimulus interval $\Delta T = t_{i,j+1} - t_{i,j}$ between subsequent sequence elements $\zeta_{i,j}$ and $\zeta_{i,j+1}$ within a sequence s_i • inter-sequence time interval $\Delta T_{\text{seq}} = t_{i+1,1} - t_{i,C_i}$ between subsequent sequences s_i and s_{i+1} • example sequence sets: – sequence set I: $\mathcal{S} = \{\{A,F,B,D\}, \{A,F,C,E\}\}$ – sequence set II: $\mathcal{S} = \{\{A,F,B,D\}, \{A,F,C,E\}, \{A,F,G,H\}, \{A,F,I,J\}, \{A,F,K,L\}\}$ • sequence i of length C_i is generated by uniformly and independently drawing C_i elements from a vocabulary of A unique token with equal probability
Output
• somatic spike times $\{t_i^k \forall i \in \mathcal{E}, k = 1, 2, \dots\}$ • dendritic currents $I_{ED,i}(t)$ ($\forall i \in \mathcal{E}$)

Table 7: Description of the input and the output. Parameter values are given in Tab. 9.

Initial conditions and network realizations
• membrane potentials: $V_i(0) = V_r$ ($\forall i \in \mathcal{E} \cup \mathcal{I}$) • dendritic currents: $I_{ED,i}(0) = 0$ ($\forall i \in \mathcal{E}$) • external currents: $I_{EX}(0) = 0$ ($\forall i \in \mathcal{E}$) • inhibitory currents: $I_{EI,i}(0) = 0$ ($\forall i \in \mathcal{E}$) • excitatory currents: $I_{IE,i}(0) = 0$ ($\forall i \in \mathcal{I}$) • synaptic permanences: $P_{ij}(0) \sim \mathcal{U}(P_{0,\min}, P_{0,\max})$ (uniform distribution; $\forall i, j \in \mathcal{E}$) • minimum permanence: $P_{\min,ij} = P_{ij}(0)$ • synaptic weights: $J_{ij}(0) = 0$ ($\forall i, j \in \mathcal{E}$) • spike traces: $x_i(0) = 0$ ($\forall i \in \mathcal{E}$) • connectivity and initial weights are randomly and independently drawn for each network realization
Simulation details
• network simulations performed in NEST (Gewaltig & Diesmann, 2007) version 3.6 (Villamar et al., 2023). • definition of excitatory neuron model and plastic synapse model using NESTML (Plotnikov et al., 2016; Linssen et al., 2025) version 8.0.0 (Linssen et al., 2024). • synchronous update using exact integration of system dynamics on discrete-time grid with step size Δt Rotter & Diesmann (1999)

Table 8: Description of the initial conditions and simulation details. Parameter values are given in Tab. 9.

1.1 Model and simulation parameters

Name	Value	Description
Network		
N_E	6240	number of excitatory neurons
N_I	26	number of inhibitory neurons
M	26	number of subpopulations
n_E, n_I	240, 1	number of excitatory and inhibitory neurons per subpopulation
ρ	20	(target) number of active neurons per subpopulation after learning = minimal number of coincident excitatory inputs required to trigger a spike in postsynaptic inhibitory neurons
N_X	26	number of external spike sources
(Potential) Connectivity		
K_{EE}	936	number of excitatory inputs per excitatory neuron (EE in-degree)
p	$K_{EE}/N_E = 0.15$	connection probability
K_{EI}	1	number of inhibitory inputs per excitatory neuron (EI in-degree)
K_{IE}	n_E	number of excitatory inputs per inhibitory neuron (IE in-degree)
K_{II}	0	number of inhibitory inputs per inhibitory neuron (II in-degree)
Excitatory neurons		
$\tau_{m,E}$	10 ms	membrane time constant
$\tau_{ref,E}$	10 ms	absolute refractory period
C_m	250 pF	membrane capacity
V_r	0.0 mV	reset potential
θ_E	20 mV	somatic spike threshold
I_{dAP}	200 pA	dAP current plateau amplitude
τ_{dAP}	60 ms	dAP duration
θ_{dAP}	59 pA	dAP threshold
Inhibitory neurons		
$\tau_{m,I}$	5 ms	membrane time constant
$\tau_{ref,I}$	2 ms	absolute refractory period
C_m	250 pF	membrane capacity
V_r	0.0 mV	reset potential
θ_I	15 mV	spike threshold

Table 9: Model and simulation parameters (continued on next page).

Name	Value	Description
Synapse		
J_{IE}	~ 581.19 pA	weight of IE connections (EPSC amplitude)
J_{EI}	~ -12915.49 pA	weight of EI connections (IPSC amplitude)
J_{EX}	~ 4112.20 pA	weight of EX connections (EPSC amplitude)
τ_{EE}	5 ms	synaptic time constant of EE connections
τ_{EI}	1 ms	synaptic time constant of EI connections
τ_{EX}	2 ms	synaptic time constant of EX connection
τ_{IE}	0.5 ms	synaptic time constant of IE connections
d_{EE}	2 ms	delay of EE connections (dendritic)
d_{IE}	0.1 ms	delay of IE connections
d_{EI}	0.1 ms	delay of EI connections
d_{EX}	0.1 ms	delay of EX connections
Plasticity		
λ_+	0.6	potentiation rate
λ_-	{0.1, 0.8}	depression rate
P_θ	10	synapse maturity threshold
$P_{\max}$	20	permanence upper bound
$P_{0,\max}$	8	permanence initialization upper bound
$P_{0,\min}$	0	permanence initialization lower bound
$J_{\max}$	12.98 pA	maximum weight
τ_+	20 ms	plasticity time constant (potentiation)
τ_-	20 ms	plasticity time constant (depression)
τ_P	80 s	permanence leak time constant
$\Delta t_{\min}$	4 ms	minimum time lag between pairs of pre- and postsynaptic spikes at which synapses are potentiated given the spike trace of the presynaptic neuron of the current time step
$\Delta t_{\max}$	50 ms	maximum time lag between pairs of pre- and postsynaptic spikes at which synapses are potentiated
Input		
S	1	number of sequences per sequence set
C_i	[10, 150]	number of characters per sequence
A	26	alphabet length
ΔT	50 ms	inter-stimulus interval (during training)
ΔT_{seq}	$\mathcal{U}(100, 105)$ ms (uniform distribution)	inter-sequence interval
Simulation		
Δt	0.1 ms	time resolution

Table 9: Model and simulation parameters.

References

- Gewaltig, Marc-Oliver and Diesmann, Markus Gewaltig, M.-O., & Diesmann, M. (2007). NEST (NEural Simulation Tool). ScholarpediaJ, 4, 1430, <https://doi.org/10.4249/scholarpedia.1430>
- Villamar, J., Vogelsang, J., Linssen, C., Kunkel, S., Kurth, A., Schöfmann, C. M., Benelhedi, M. A., Babu, P. N., Eppler, J. M., de Schepper, R., Mitchell, J., Morrison, A., Haug, N., Diaz, S., Acimovic, J., Graber, S., Jiang, H.-J., Terhorst, D., Spreizer, S., Welle Skaar, J.-E., Stapmanns, J., Manninen, T., Krüger, M., Lehtimäki, M., Ito, S., Lee, A. Y., Lindahl, M. & Plesser, H. E. NEST 3.6
- Plotnikov, D., Blundell, I., Ippen, T., Eppler, J. M., Rumpe, B., Morrison, A. NESTML: a modeling language for spiking neurons. Modellierung 2016, P-254, pages 93-108
- Linssen, C., Babu, P. N., Eppler, J. M., Koll, L., Rumpe, B., Morrison, A. NESTML: a generic modeling language and code generation tool for the simulation of spiking neural networks with advanced plasticity rules. Frontiers Neuroinformatics, 19, 10.3389/fninf.2025.1544143
- Linssen, C., Bouhadjar, Y., Ewert, L., Lober, M., Babu, P., Feller, F., Wybo, W., Morrison, A., Rumpe, B. NESTML (v8. 0.0)

Rotter, S., & Diesmann, M. (1999). Exact digital simulation of time-invariant linear systems with applications to neuronal modeling. *Biol. Cybern.* 81(5-6), 381–402.